

Introduction

The OSI model (or Open Systems Interconnection Model) is a fundamental model used in networking. This model provides a framework explaining how all networked devices will send, receive and interpret data.

One of the main benefits of the OSI model is that devices can have different functions and designs on a network while communicating with other devices. Data sent across a network that follows the uniformity of the OSI model can be understood by other devices.

The OSI model consists of seven layers which are described below. Each layer has a different set of responsibilities and is arranged from Layer 7 to Layer 1.

Some models contain Layer 8 which is considered to be the User or Political layer (Human Layer).

At every individual layer that data travels through, specific processes take place, and pieces of information are added to this data.

The Application Layer (7)

The application layer of the OSI model is the layer that you will be most familiar with. This familiarity is because the application layer is the layer in which protocols and rules are in place to determine how applications interact with data sent or received by the user.

Everyday applications such as email clients, browsers, or file server browsing software such as FileZilla provide a friendly, Graphical User Interface (GUI) for users to interact with data sent or received. Application layer protocols include HTTP as well as SMTP (Simple Mail Transfer Protocol is one of the protocols that enables email communications).

The Presentation Layer (6)

Layer 6 of the OSI model is the layer in which translation, encoding, and formatting happen. Because software developers can develop any software such as an email client differently, the data still needs to be handled in the same way — no matter how the software works.

This layer acts as a translator for data to and from the application layer (layer 7). The receiving computer will also understand data sent to a computer in one format destined for in another format. For example, when you send an email, the other user may have another email client to you, but the contents of the email will still need to display the same.

The Session Layer (5)

Once data has been correctly translated or formatted from the presentation layer (layer 6), the session layer (layer 5) will begin to create a connection to the other computer that the data is destined for. When a connection is established, a session is created. Whilst this connection is active, so is the session.

The session layer (layer 5) synchronises the two computers to ensure that they are on the same page before data is sent and received. Checkpoints are periodically put in place in case connection is lost. Once these checks are in place, the session could be resumed from the last checkpoint. Without the checkpoints, the entire transfer would have to begin again from the start (think of it as loading a save file in a video game).

The Transport Layer (4)

Layer 4 of the OSI model plays a part in transmitting data across a network. When data is sent between devices, it follows one of two different protocols that are decided based upon several factors:

The Transmission Control Protocol (TCP). This protocol is designed to be reliable and guaranteed. TCP maintains state about the connection, such as who initiated it and what stage of the handshake or teardown it's in. TCP also incorporates error checking. This is how TCP can guarantee that data sent from the small chunks in the session layer (layer 5) has then been received and reassembled in the same order.

- Advantages of TCP
 - Guarantees the accuracy of data.
 - Synchronises two devices to prevent flooding of data.
 - Performs a lot more processes for reliability.
- Disadvantages of TCP
 - Requires a reliable connection between the two devices. If one small chunk of data is not received, then the entire chunk of data cannot be used.
 - A slow connection can bottleneck another device as the connection will be reserved on the receiving computer the whole time.
 - TCP is significantly slower than UDP because more work has to be done by the devices using this protocol.

The User Datagram Protocol (UDP). This protocol is not nearly as advanced as the TCP protocol. There is no synchronisation between the two devices or guarantee. Whilst this sounds disadvantageous, it does have its merits:

- Advantages of UDP
 - UDP is much faster than TCP.
 - UDP leaves the application layer (user software) to decide if there is any control over how quickly packets are sent.
 - UDP does not reserve a continuous connection on a device as TCP does.
- Disadvantages of UDP
 - UDP doesn't care if the data is received.
 - It is quite flexible to software developers in this sense.
 - This means that unstable connections result in a terrible experience for the user.

UDP is useful in situations where timeliness matters more than reliability. For example, larger files such as video streaming or VOIP (where it is okay if some parts of the data are lost. Pixelated video or choppy audio is okay here and there.)

The Network Layer (3)

The network layer is where the routing of data takes place. Firstly, routing simply determines the most optimal path in which these chunks of data should be sent.

Whilst some protocols at this layer determine exactly what is the "optimal" path that data should take to reach a device, we should only know about their existence at this stage of the networking module. Briefly, these protocols include OSPF (Open Shortest Path First) and RIP (Routing Information Protocol). The factors that decide what route is taken is decided by the following:

What path is the shortest? I.e. has the least amount of devices that the packet needs to travel across.

What path is the most reliable? I.e. have packets been lost on that path before?

Which path has the faster physical connection? I.e. is one path using a copper connection (slower) or a fibre (considerably faster)?

At this layer, everything is dealt with via IP addresses such as 192.168.1.100. Devices such as routers capable of delivering packets using IP addresses are known as Layer 3 devices — because they are capable of working at the third layer of the OSI model.

The Data Link Layer (2)

The data link layer focuses on the physical addressing of the transmission. It receives a packet from the network layer (including the IP address for the remote computer) and adds in the physical MAC (Media Access Control) address of the receiving endpoint. Inside every network-enabled computer is a Network Interface Card (NIC) which comes with a unique MAC address to identify it.

MAC addresses are set by the manufacturer and literally burnt into the card; You normally **can't permanently change them**, only **spoof** them in the OS. When information is sent across a network, it's actually the physical address that is used to identify where exactly to send the information.

Additionally, it's also the job of the data link layer to present the data in a format suitable for transmission.

The Physical Layer (1)

This layer is one of the easiest layers to grasp. Put simply, this layer references the physical components of the hardware used in networking and is the lowest layer that you will find. Devices use electrical signals to transfer data between each other in a binary numbering system (1's and 0's).

For example, ethernet cables connecting devices.